

Mathematical Foundations for Symmetric Programming

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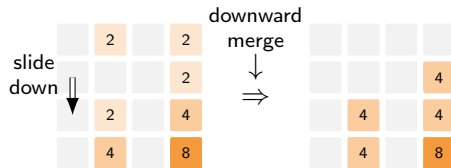
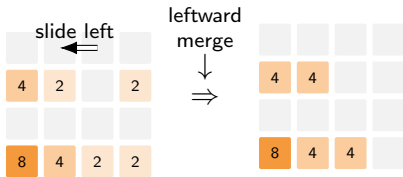


Advanced
Research
+ Innovation
Agency

ARIA

Goal: exploit **symmetry** for **computation** and **reasoning**

Example: 2048 game¹ [Cirulli'14]



¹Available to play: <https://play2048.co>

2048 in Idris 2 [Brady'21]

Merging left goes with the grain

```
Row, Board : Nat -> Type
Row  n = Vect n Int
Board n = Vect n (Row n)

(.pad0) : Row n -> Row (1 + n)
xs.pad0 = replace {p = Row}
            (plusCommutative _ _)
            $ xs ++ [0]

mergeRow : Row n -> Row n
mergeRow [] = []
mergeRow ( 0 :: xs) = (mergeRow xs).pad0
mergeRow ( x :: []) = [x]
mergeRow (x :: 0 :: xs) = (mergeRow (x :: xs)).pad0
mergeRow (x :: y :: xs) with (x == y)
  _ | True  = 2*x :: (mergeRow xs).pad0
  _ | False =  x :: mergeRow (y :: xs)

mergeLeft : Board n -> Board n
mergeLeft = map mergeRow
```

Merging in 2048

Going against the grain

merging
up/down
incompatible
with inductive
structure



use a different
representation



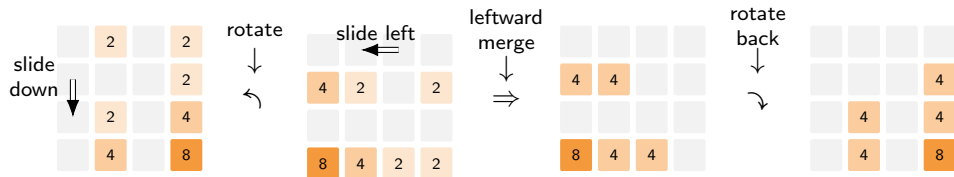
parameters
galore!²

```
var vector      = this.getVector(direction);
var traversals = this.buildTraversals(vector);
// ...
GameManager.prototype.getVector =
function (direction) { // Vectors representing tile movement
  var map = { 0: { x: 0, y: -1 }, // Up
              1: { x: 1, y: 0 },  // Right
              2: { x: 0, y: 1 },  // Down
              3: { x: -1, y: 0 } }; // Left
  return map[direction];
};
```

²See the function `GameManager.prototype.move`:

https://github.com/gabrielecirulli/2048/blob/478b6ec346e3787f589e4af751378d06ded4cbbc/js/game_manager.js#L129.

Easy 2048 implementation using **symmetry**

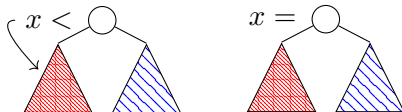


Anecdotal evidence with half-a-dozen UG students

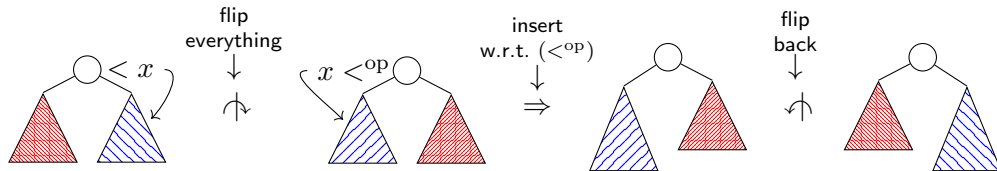
- ✗ buggy parametric solution
- ✓ mostly right symmetric solution

Example: binary search tree insertion using **symmetry**

Interesting cases



Symmetric case



NB: Yes, silly performance. Won't improve performance today, more later.

Example: **proofs** using **symmetry** [adapted from Harrison'09]

Theorem (Schur's inequality)

Then, for all $x_1, x_2, x_3 > 0$, we have, for all t :

$$x_1^t(x_1 - x_2)(x_1 - x_3) + x_2^t(x_2 - x_3)(x_2 - x_1) + x_3^t(x_3 - x_1)(x_3 - x_2) \geq 0$$

Proof

Without loss of generality, assume $x_1 \geq x_2 \geq x_3$. Then:

$$\begin{aligned} x_1^t(x_1 - x_2)(x_1 - x_3) + x_2^t(x_2 - x_3)(x_2 - x_1) + x_3^t(x_3 - x_1)(x_3 - x_2) &= \\ (x_1 - x_2) \left(x_1^t(x_1 - x_3) - x_2^t(x_2 - x_3) \right) + x_3^t(x_1 - x_3)(x_2 - x_3) &\geq \\ (x_1 - x_2)(x_1^t - x_2^t)(x_1 - x_3) + x_3^t(x_1 - x_3)(x_2 - x_3) &\geq 0 \end{aligned}$$



Example: **proofs** using **symmetry** [adapted from Harrison'09]

Theorem (Schur's inequality)

Then, for all $x_1, x_2, x_3 > 0$, we have, for all t :

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Rocq SSReflect³

The general syntax of without loss is:

```
wlog suff? clear_switch? i_item? : ident* / term
```

On a goal **G**, it creates two subgoals: a first one to prove the formula $(\text{term} \rightarrow \mathbf{G}) \rightarrow \mathbf{G}$ and a second one to prove the formula $\text{term} \rightarrow \mathbf{G}$. If the optional list of `ident` is present on the left side of `/`, these constants are generalized in the premise $(\text{term} \rightarrow \mathbf{G})$ of the first subgoal.

³Rocq Reference Manual:

<https://rocq-prover.org/doc/V9.2.0/refman/proof-engine/ssreflect-proof-language.html#rocq:tacn.wlog>

Agda⁴

```
-- Without Loss of Generality
```

```
module _ {_R_ : Rel A ℓ1} {Q : Rel A ℓ2} where
```

```
wlog : Total _R_ → Symmetric Q →  
  (∀ a b → a R b → Q a b) →  
  ∀ a b → Q a b
```

⁴Agda Stdlib:

<https://github.com/agda/agda-stdlib/blob/c2abce1b8e72ccae49f73791d0996eaf23674357/src/Relation/Binary/Consequences.agda#L197>

Lean4⁵

The tactic `wlog h : P` will add an assumption `h : P` to the main goal, and add a new goal that requires showing that the case $\neg h : P$ can be reduced to the case where `P` holds (typically by symmetry).

⁵Mathlib4 documentation:

https://leanprover-community.github.io/mathlib4_docs/Mathlib/Tactic/WLOG.html

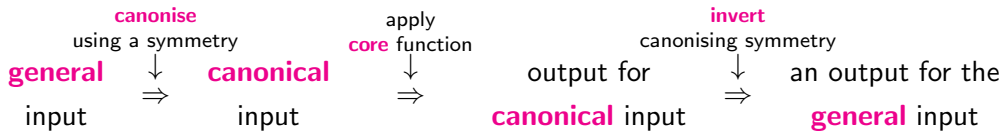
uniform w.l.o.g. using symmetry

$$\begin{array}{ccccccc} & \text{sort using } \pi & & \text{prove using} & & \text{permute back} & \\ & \downarrow & & \downarrow & & \downarrow & \\ x_1, x_2, x_3 \in \mathbb{R}_+ & \Rightarrow & x_{\pi 1} \geq x_{\pi 2} \geq x_{\pi 3} & \Rightarrow & P(x_{\pi 1}, x_{\pi 2}, x_{\pi 3}) & \Rightarrow & P(x_1, x_2, x_3) \\ & & \text{additional assumption} & & \text{since } P \text{ is invariant} & & \end{array}$$

NB: revisionism:

- ✓ Dijkstra'95 [EWD1223, not mechanised]
- ✓ HOL WLOG tactics [Harrison'09]
- ✗ Rocq SSReflect
- ✓ Agda Stdlib
- ✗ Lean

w.l.o.g. = (input & output **symmetries**) + (**canonising** symmetries) + (**core** function/proof)



symmetry = **group** + **action**

Symmetric programming

Exploit symmetries for computation and reasoning—**avoid symmetric cases**.

Mathematical foundations for symmetric programming semantics:

- ▶ **Abstractions** for symmetric programming
- ▶ **Simply-typed** reasoning about equivariance
- ▶ **Dependently-typed** and mathematical reasoning
- ▶ User-defined **data-types** through initial algebra semantics and induction principles
 - ▶ Simply-typed data-types
 - ▶ Dependently-typed/indexed data-types

Talk structure

- ▶ Introduction
- ▶ Remedial group theory basics
- ▶ Simply-typed symmetric programming
- ▶ Dependently-typed symmetric programming
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Remedial group theory basics

Group action $\langle G, A \rangle$

Group $G = \langle \underline{G}, \dots \rangle$

G -action $A = \langle \underline{A}, \dots \rangle$

Intuition

- ▶ **symmetries**: $\underline{G}$ elements
- ▶ x **symmetric to** y **via** g

action and group axioms mean:

‘**being symmetric**’ is a
proof-relevant equivalence relation tracked by
symmetries

Remedial group theory basics

Group action $\langle G, A \rangle$

Group $G = \langle \underline{G}, (*), e, (-)^{-1} \rangle$:

$\underline{G}$ (**carrier set**) $\frac{g, h : \underline{G}}{g * h : \underline{G}}$ (**multiplication/composition**)

$\frac{}{e : \underline{G}}$ (**unit/identity/neutral**) $\frac{g : \underline{G}}{g^{-1} : \underline{G}}$ (**inverse**)

such that:

$g, h, k : \underline{G} \vdash (g * h) * k = g * (h * k)$ (associativity)

$g : \underline{G} \vdash g * e = g = e * g$ (neutrality)

$g : \underline{G} \vdash g * g^{-1} = e = g^{-1} * g$ (inverses)

G-action $A = \langle \underline{A}, \dots \rangle \dots$

Board rotations $\langle \mathbf{C}_4, \text{Dir} \rangle$

$\underline{\mathbf{C}}_4 := \{\hookleftarrow, \curvearrowleft, \curvearrowright, \mathbf{Q}\}$

$\hookleftarrow * \hookleftarrow := \curvearrowleft \quad \hookleftarrow * \curvearrowleft := \curvearrowright$

$\hookleftarrow * \curvearrowright := \mathbf{Q} \quad \hookleftarrow * \mathbf{Q} := \hookleftarrow$

group laws imply rest and:

$e = \mathbf{Q}$

$\hookleftarrow^{-1} = \curvearrowright \quad \curvearrowleft^{-1} = \curvearrowleft$

Remedial group theory basics

Group action $\langle G, A \rangle$

Group $G = \langle \underline{G}, (*), e, (-)^{-1} \rangle$

G-action $A = \langle \underline{A}, \overset{A}{*} \rangle :$

$$\underline{A}(\text{carrier set}) \quad \frac{g : \underline{G} \quad x : \underline{A}}{g \overset{A}{*} x : \underline{A}} (\text{action of } G \text{ on } \underline{A})$$

such that:

$$\begin{aligned} g, h : \underline{G}, x : \underline{A} \vdash g \overset{A}{*} (h \overset{A}{*} x) &= (g * h) \overset{A}{*} x && (\text{associativity}) \\ x : \underline{A} \vdash e \overset{A}{*} x &= x && (\text{neutrality}) \end{aligned}$$

Board rotations $\langle \mathbf{C}_4, \text{Dir} \rangle$

$$\underline{\mathbf{C}}_4 := \{ \swarrow, \searrow, \nearrow, \nwarrow \}$$

$$\begin{aligned} \swarrow * \swarrow &:= \searrow & \swarrow * \searrow &:= \nearrow \\ \swarrow * \nearrow &:= \nwarrow & \swarrow * \nwarrow &:= \swarrow \end{aligned}$$

acting on

$$\underline{\text{Dir}} := \{ \Leftarrow, \Rightarrow, \Uparrow, \Downarrow \}$$

$$\begin{aligned} \swarrow * \Leftarrow &:= \Downarrow & \swarrow * \Downarrow &:= \Rightarrow \\ \swarrow * \Rightarrow &:= \Uparrow & \swarrow * \Uparrow &:= \Leftarrow \end{aligned}$$

laws determine the rest
acting similarly on:

$$\text{Board}_n \text{ and } \text{Dir} \times \text{Board}_n$$

Remedial group theory basics

Group action $\langle G, A \rangle$

Group $G = \langle \underline{G}, (*), e, (-)^{-1} \rangle$

G-action $A = \langle \underline{A}, \rangle$

Notation

► **symmetries**: $\underline{G}$ elements

► x **symmetric to** y **via** g when: $x = g \circledast y$

action and group axioms mean:

‘**being symmetric**’ is a
proof-relevant equivalence relation tracked by
symmetries

Board rotations $\langle \mathbf{C}_4, \mathbf{Dir} \rangle$

$\underline{\mathbf{C}}_4 := \{ \curvearrowleft, \curvearrowright, \curvearrowright, \mathbf{Q} \}$

$\curvearrowleft * \curvearrowleft := \curvearrowright \quad \curvearrowleft * \curvearrowright := \curvearrowright$

$\curvearrowleft * \curvearrowright := \mathbf{Q} \quad \curvearrowleft * \mathbf{Q} := \curvearrowleft$

acting on

$\underline{\mathbf{Dir}} := \{ \Leftarrow, \Rightarrow, \Uparrow, \Downarrow \}$

$\curvearrowleft \circledast \Leftarrow := \Downarrow \quad \curvearrowleft \circledast \Downarrow := \Rightarrow$

$\curvearrowleft \circledast \Rightarrow := \Uparrow \quad \curvearrowleft \circledast \Uparrow := \Leftarrow$

laws determine the rest
acting similarly on:

$\mathbf{Board}_n$ and $\mathbf{Dir} \times \mathbf{Board}_n$

Symmetry-preserving functions

Equivariant map $\varphi : A \circledast \rightarrow B$ between G -actions A, B

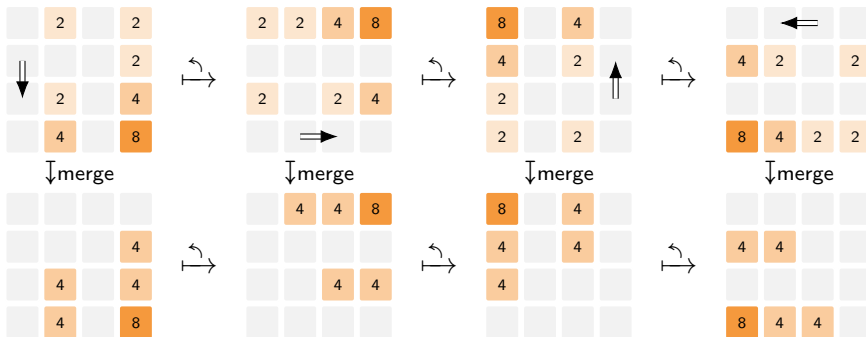
$\varphi : \underline{A} \rightarrow \underline{B}$
function

$g : \underline{G}, x : \underline{A} \vdash$

$\varphi(g \circledast x) = g \circledast (\varphi x) : \underline{B}$

i.e.: $\frac{x, y \text{ symmetric via } g}{f x, f y \text{ symmetric via } g}$

Example: $\text{merge} : \text{Dir} \times \text{Board}_n \circledast \rightarrow \text{Board}_n$



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w.l.o.g. anatomy: **canonisation**

w.l.o.g. = (input & output **symmetries**) + (**canonising** symmetries) + (**core** function/proof)

Canoniser for G -action A

function $c : \underline{A} \rightarrow \underline{G}$ assigning **canonising symmetries** to A -elements.

Canonical elements

$$\text{Can } c := \{ (c a) \otimes a \mid a \in \underline{A} \}$$

Example: $\text{mkLft} : \underline{\text{Dir}} \rightarrow \underline{\text{C}}_4$

Canonising symmetry makes direction 'left':

$$\text{mkLft} \Rightarrow := \curvearrowleft$$

$$\text{mkLft} \Uparrow := \curvearrowright$$

$$\text{mkLft} \Leftarrow := \curvearrowleft$$

$$\text{mkLft} \Downarrow := \curvearrowright$$

Only canonical direction is 'left':

$$\text{Can } \text{mkLft} = \{ \Leftarrow \}$$

Lemma

$$\text{Can} \left(\underline{A} \times \underline{B} \xrightarrow{\pi_1} \underline{A} \xrightarrow{c} \underline{G} \right) = \text{Can} \left(\underline{A} \xrightarrow{c} \underline{G} \right) \times \underline{B}$$

Example

Canonise merge-inputs by canonising the direction:

$$c : \underline{\text{Dir}} \times \underline{\text{Board}}_n \xrightarrow{\pi_1} \underline{\text{Dir}} \xrightarrow{\text{mkLft}} \underline{\mathbf{C}}_4$$

By this Lemma, in canonical merge-inputs, merge-direction left but board unrestricted:

$$\text{Can } c = \{ \Leftarrow \} \times \underline{\text{Board}}_n$$

w.l.o.g. = (input & output **symmetries**) + (**canonising** symmetries) + (**core** function/proof)

Core function $f : \text{Can } c \rightarrow \underline{B}$

work on canonical elements, typically more concrete, after eliminating symmetric cases

Example: merging left

$\text{merge}_{\leftarrow}^{\text{row}} : \text{Row}_n \rightarrow \text{Board}_n$

$\text{merge}_{\leftarrow}^{\text{row}} \langle \rangle := \langle \rangle$

$\text{merge}_{\leftarrow}^{\text{row}} \langle 0 \rangle \mathbin{++} r := (\text{merge}_{\leftarrow}^{\text{row}} r) \mathbin{++} \langle 0 \rangle$

$\text{merge}_{\leftarrow}^{\text{row}} \langle x \rangle := \langle x \rangle$

$\text{merge}_{\leftarrow}^{\text{row}} (\langle x, 0 \rangle \mathbin{++} r) := \text{merge}_{\leftarrow} (\langle x \rangle \mathbin{++} r) \mathbin{++} \langle 0 \rangle$

$\text{merge}_{\leftarrow}^{\text{row}} (\langle 2^k, 2^k \rangle \mathbin{++} r) := \langle 2^{k+1} \rangle \mathbin{++} (\text{merge}_{\leftarrow}^{\text{row}} r) \mathbin{++} \langle 0 \rangle$

$\text{merge}_{\leftarrow}^{\text{row}} (\langle 2^k, 2^\ell \rangle \mathbin{++} r) := \langle 2^k \rangle \mathbin{++} \text{merge}_{\leftarrow}^{\text{row}} (\langle 2^\ell \rangle \mathbin{++} r)$

$\text{merge}_{\leftarrow} : \text{Board}_n \rightarrow \text{Board}_n$

$\text{merge}_{\leftarrow} := \text{map } \text{merge}_{\leftarrow}^{\text{row}}$

Symmetric programming **abstraction**

w.l.o.g. = (input & output **symmetries**)+(**canonising** symmetries)+(**core** function/proof)

W.l.o.g. construction

Formally:

$$\frac{G\text{-actions } A, B \quad c : \underline{A} \rightarrow \underline{G} \quad f : \text{Can } c \rightarrow \underline{B}}{\text{wlog}(c, f) := \left(a \mapsto (ca)^{-1} \circledast f((ca) \circledast a) \right) : \underline{A} \rightarrow \underline{B}}$$

Example

```
mergeUnchecked : {n : Nat} -> Dir -> Board n -> Board n
```

```
mergeUnchecked = Prelude.curry $ wlog
```

```
  (\(dir, _) => makeLeft dir)           -- canoniser
  (\case (Lt, board) => mergeLeft board  -- core function
    _ => believe_me "oops")             -- unreachable case
```

Symmetric programming **abstraction**

w.l.o.g. = (input & output **symmetries**) + (**canonising** symmetries) + (**core** function/proof)

W.l.o.g. construction

Formally:

$$\frac{G\text{-actions } A, B \quad c : \underline{A} \rightarrow \underline{G} \quad f : \text{Can } c \rightarrow \underline{B}}{\text{wlog}(c, f) := \left(a \mapsto (ca)^{-1} \otimes f((ca) \otimes a) \right) : \underline{A} \rightarrow \underline{B}}$$

Example

```
core : {n : Nat} -> Canon (Canoniser {n}) -> Board n
```

```
core ((d, b) ** ev) with 0 (canonicalView ev)  
  core ((Lt, b) ** ev) | Refl = mergeLeft b
```

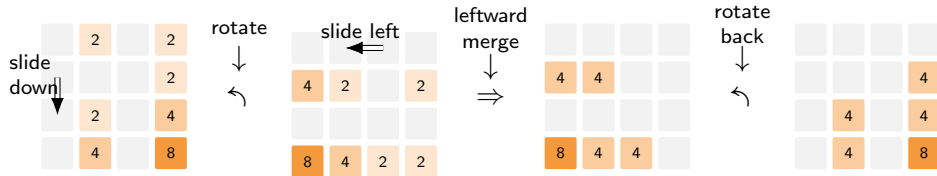
```
merge : {n : Nat} -> Dir -> Board n -> Board n
```

```
merge = Prelude.curry $ wlog' Canoniser core
```

Symmetric programming in Idris 2 [Brady'21]

Simply-typed w.l.o.g.

- ▶ Straightforward to implement
- ▶ Straightforward to use
- ▶ Plumbing group actions with type-classes/interfaces fiddly as usual
- ▶ Plenty of 'believe_me' in code for unreachable cases
- ▶ Doesn't cover mathematical reasoning



TwentyFortyEight.idr

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Back to mathematics

these concepts apply to **propositions** & **proofs**

concepts apply **uniformly** via dependent types:

types as propositions; programs as proofs



G-actions on propositions; **canonisers** and **core functions** in **wlog** proofs

easy with **simple types** (HOL/FOL), **duplicating** concepts in different **kinds/stages**
(terms vs. propositions, object-level vs. meta-level)

analyse extant w.l.o.g. principles of Dijkstra & Harrison

Digression: mathematics with dependent types

keep meta-language **implicit**:

mathematical foundations have no room in polite mathematical discourse

instead, use concepts that **explicate** type dependency:

indexed families and **terms-in-context**

analogy: vernacular use of sets, functions, relations, etc.

without necessarily committing to ZF, nor requiring moving to a meta-level

one can mould families into **various models** of **various type theories**:

I don't want to, it adds **little value** and **much noise** at the mathematical level

formalisation and esp. **mechanisation** add real value,

but cement a specific (formal) meta-language—left for **future work**

Explicating type dependency

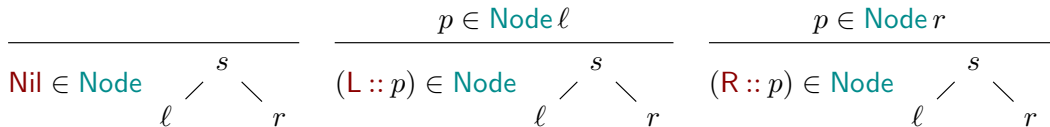
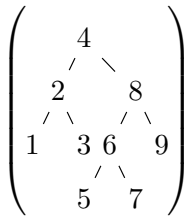
Family $\Gamma \vdash X$

- ▶ Γ set of **indices**
- ▶ X a family assigning a set X_γ to each index $\gamma \in \Gamma$ —the **fibre** over γ

Example: internal tree nodes $t : \text{Tree } S \vdash \text{Node } t$

- ▶ index $\text{Tree } S \ni t$ binary tree with labels in S ; $R :: L :: \text{Nil} \in \text{Node}$
- ▶ fibre $\text{Node } t$: paths ending in internal node of $\text{Tree } S \ni t$.

Fibres collectively form the smallest sets of lists over $\{L, R\}$ containing:



Explicating type dependency

Term/fibred element $\Gamma \vdash M : X$

- ▶ family $\Gamma \vdash X$
- ▶ M a sequence assigning an element $M_\gamma \in X_\gamma$ at every fibre over $\gamma \in \Gamma$

Example

$$\begin{array}{l} t, s : \text{Tree} \mathbb{N} \vdash \\ R :: L :: \text{Nil} : \text{Node} \end{array} \left(\begin{array}{c} 4 \\ / \quad \backslash \\ 2 \quad 8 \\ / \quad \backslash \quad / \quad \backslash \\ 1 \quad 3 \quad 6 \quad 9 \\ \quad / \quad \backslash \\ \quad t \quad s \end{array} \right)$$

$$\begin{array}{l} \text{element at fibre } t \mapsto 5, s \mapsto 7: \\ R :: L :: \text{Nil} \in \text{Node} \end{array} \left(\begin{array}{c} 4 \\ / \quad \backslash \\ 2 \quad 8 \\ / \quad \backslash \quad / \quad \backslash \\ 1 \quad 3 \quad 6 \quad 9 \\ \quad / \quad \backslash \\ \quad 5 \quad 7 \end{array} \right)$$

Formalising type dependency

Straightforward⁶ abstractions for type dependency:

$$\begin{array}{c}
 \frac{\Gamma \vdash X}{\Gamma, x : X \text{ set}} \quad \frac{\theta : \Gamma \rightarrow \Delta \quad \Delta \vdash Y}{\Gamma \vdash Y [\theta]} \quad \frac{\Gamma \vdash X}{\text{wkn} : (\Gamma, x : X) \rightarrow \Gamma} \quad \frac{\Gamma \vdash X}{\Gamma, x : X \vdash x : X [\text{wkn}]} \\
 \\
 \frac{\theta : \Delta \rightarrow \Gamma \quad \Delta \vdash M : X [\theta]}{(\theta, x \mapsto M) : \Delta \rightarrow (\Gamma, x : X)} \quad \frac{\Gamma \vdash X \quad \Gamma, x : X \vdash Y}{\Gamma \vdash (x : X) \times Y} \quad \frac{\Gamma \vdash M : X \quad \Gamma \vdash K : Y [x \mapsto M]}{\Gamma \vdash M, K : (x : X) \times Y} \\
 \\
 \frac{\Gamma \vdash M : (x : X) \times Y}{\Gamma \vdash M.\text{fst} : X} \quad \frac{\Gamma \vdash M : (x : X) \times Y}{\Gamma \vdash M.\text{snd} : Y [x \mapsto M.\text{fst}]} \quad \frac{\Gamma \vdash X \quad \Gamma, x : X \vdash Y}{\Gamma \vdash (X : x) \rightarrow Y} \\
 \\
 \frac{\Gamma \vdash K : (X : x) \rightarrow Y \quad \Gamma \vdash M : X}{\Gamma \vdash K M : Y [x \mapsto M]} \quad \frac{\Gamma, x : X \vdash M : Y}{\Gamma \vdash (\lambda x : X. M) : (x : X) \rightarrow Y}
 \end{array}$$

⁶By 'straightforward', I mean: experienced type theorists 'do the usual thing'. Inexperienced theorists need to learn 'the usual things', so working it out may take some time and guidance.

Formalising type dependency

Straightforward⁷ because defer type-dependency difficulties to meta-level:

- ▶ Dependent-pairs/Grothendieck construction $\coprod_{\Gamma} X := \{\gamma, x \mid \gamma \in \Gamma, x \in X_{\gamma}\}$
- ▶ Dependent-functions/products $\prod_{\Gamma} X$

Can turn these into a **formal** or even **mechanised** type theory, and mould families into your **favourite model** of this type-theory, including a **shallow embedding** if your metalanguage is dependently-typed.

As usual, your mileage may vary, and independently of this development.
(So that's on you, and we can discuss offline.)

End of digression, returning to **dependently-typed symmetric programming**

⁷By 'straightforward', I mean: experienced type theorists 'do the usual thing'. Inexperienced theorists need to learn 'the usual things', so working it out may take some time and guidance.

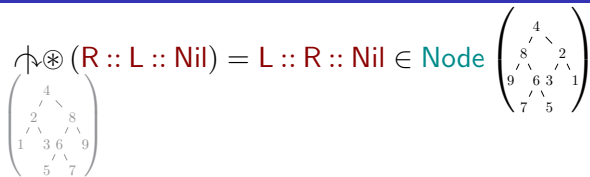
Dependently-typed actions

Indexed G -action $\Gamma \vdash A$

► $\Gamma = \langle \underline{\Gamma}, (\otimes) \rangle$ **indexing** G -action

► $\underline{\Gamma} \vdash \underline{A}$ **carrier** family

► $\frac{\vdash g : \underline{G} \quad \gamma : \Gamma \vdash a : \underline{A}}{\gamma : \Gamma \vdash g \otimes_{\gamma} a : \underline{A} [\gamma \mapsto g \otimes \gamma]} \quad \Gamma\text{-indexed action of } G \text{ on } \underline{A}$



such that

equivalently:

$$(\otimes) : (g : \underline{G}) \rightarrow (\gamma : \Gamma) \rightarrow \underline{A}_{\gamma} \rightarrow \underline{A}_{g \otimes \gamma}$$

$$\gamma : \Gamma, x : \underline{A} \vdash e \otimes_{\gamma} x = x \quad g, k : \underline{G}, \gamma : \Gamma, x : \underline{A} \vdash (g * k) \otimes_{\gamma} x = g \otimes_{k \otimes \gamma} (k \otimes_{\gamma} x)$$

Proposition

Indexed actions amount to equivariant projections from the Grothendieck construction:

$$\begin{array}{ccc} \Gamma \vdash A & \iff & \coprod_{\Gamma} A = (\coprod_{\underline{\Gamma}} \underline{A}, ((\otimes), (\otimes))) \\ \text{indexed } G\text{-action} & & G\text{-action} \end{array} \quad \pi_1 : \coprod_{\Gamma} A \multimap \Gamma$$

Indexed w.l.o.g. anatomy: canonisers

Indexed canoniser for G -action $\Gamma \vdash A$

term $\underline{\Gamma} \vdash c : \underline{A} \rightarrow \underline{G}$

induced **family** $\underline{\Gamma} \vdash \text{Can } c$ of canonical elements:

$$(\text{Can } c)_\gamma := \left\{ g \circledast a \middle| g \in \underline{G}, a \in \underline{A}_{g^{-1} \circledast \gamma}, c_{g^{-1} \circledast \gamma} a = g \right\}$$

Example

A canoniser for paths indexed by binary trees with action of $\mathbf{S}_2 := \{\mathbf{I}, \swarrow\}$:

$$t : \text{Tree } \mathbb{N} \vdash c : \text{Node } t \rightarrow \mathbf{S}_2 \quad c_{\text{Node}\langle \ell, n, r \rangle} \text{Nil} := \mathbf{I} =: c_{\text{Node}\langle \ell, n, r \rangle} \mathbf{L} \quad c_{\text{Node}\langle \ell, n, r \rangle} \mathbf{R} := \swarrow$$

Canonical paths may only turn **R** after first step:

$$(\text{Can } c)_{\text{Node}\langle \ell, n, r \rangle} = \{\text{Nil}\} \cup \{\mathbf{L} :: p \mid p \in \text{Node } \ell\}$$

Indexed w.l.o.g. anatomy: core functions

Indexed core function

term $\underline{\Gamma} \vdash f : (\mathbf{x} : \text{Can } c) \rightarrow \underline{B}_{\text{coerce}_{\text{Can } c \subseteq \underline{A}}} \mathbf{x}$

Example

core function returning node pointed at by a path:

$$\begin{aligned} \mathbf{t} : \text{Tree } \mathbb{N} \vdash \text{lookup}^{\text{Core}} : \text{Can } c \rightarrow \mathbb{N} \quad \text{lookup}^{\text{Core}}_{\text{Node}\langle \ell, m, r \rangle} \text{Nil} &:= m \\ \text{lookup}^{\text{Core}}_{\text{Node}\langle \ell, m, r \rangle} (\mathbf{L} :: p) &:= \text{lookup}_{\ell} p \end{aligned}$$

- ✓ **eliminated** symmetric case
- ✗ **recursive** call to lookup, not lookup^{Core}
revisit through **data-types**

Symmetric programming **dependently-typed** abstraction

indexed w.l.o.g. = (**indexed** symmetries) + (**indexed** canoniser) + (**indexed** core function)

W.l.o.g. construction

Formally:

$$\frac{G\text{-actions } \Gamma \vdash A \quad \Gamma, x : A \vdash B \quad \Gamma \vdash c : \underline{A} \rightarrow \underline{G} \quad \Gamma \vdash f : \text{Can } c \rightarrow \underline{B}}{\gamma : \underline{\Gamma} \vdash \text{wlog}(c, f) := \left(\lambda a. (ca)^{-1} \underset{(ca) \circledast \gamma}{\circledast} f \left((ca) \underset{\gamma}{\circledast} a \right) \right) : (x : \underline{A}) \rightarrow \underline{B}}$$

Example (revisit later)

$$t : \text{Tree } \mathbb{N} \vdash \text{lookup} := \text{wlog}(c, \text{lookup}^{\text{Core}}) : \text{Node } t \rightarrow \mathbb{N}$$

Talk structure

- ▶ Introduction
- ▶ Remedial group theory basics
- ▶ Simply-typed symmetric programming
- ▶ **Dependently-typed symmetric programming**
 - ▶ Indexed actions and w.l.o.g.
 - ▶ **Mathematical reasoning: Dijkstra'95 & Harrison'09**
- ▶ Symmetric user-defined data-types
 - ▶ Simply-typed data-types
 - ▶ Indexed data-types
- ▶ Equivariant reasoning

Mathematical reasoning w.l.o.g.

Propositions as types, proofs as programs

For this part, take propositions $\varphi \in \mathbf{Prop} := \{\mathbf{True}, \mathbf{False}\}$. Use trivial action as needed.

$\mathbf{Prop}$ acts as a universe with:

$$\frac{\Gamma \vdash \varphi : \mathbf{Prop}}{\Gamma \vdash \varphi} \quad \varphi_\gamma := \begin{cases} \varphi_\gamma = \mathbf{True} : & \mathbb{1} \\ \text{otherwise} : & \emptyset \end{cases}$$

$$\begin{aligned} \text{E.g.: } (x, y, z) : (0, \infty)^3 \vdash \text{Schur} := \\ \forall t : \mathbb{R}. x_1^t (x_1 - x_2)(x_1 - x_3) \\ + x_2^t (x_2 - x_3)(x_2 - x_1) \\ + x_3^t (x_3 - x_1)(x_3 - x_2) \\ \geq 0 : \mathbf{Prop} \end{aligned}$$

At most one term $\Gamma \vdash M : \varphi$. It exists iff φ holds in every fibre $\gamma \in \Gamma$.

Equivariant terms $\Gamma \vdash M : A$ (where $\Gamma \vdash A$ indexed G -action)

terms $\underline{\Gamma} \vdash M : \underline{M}$ satisfying: $\underline{g} : \underline{G}, \gamma : \underline{\Gamma} \vdash g \underset{\gamma}{\circledast} M_\gamma = M_{\underline{g} \circledast \gamma} : A_{\underline{g} \circledast \gamma}$

Proposition

$\Gamma \vdash \varphi : \mathbf{Prop}$ **invariant**
i.e., equivariant w.r.t. trivial action,



$\Gamma \vdash \varphi$ has an indexed action
(this indexed action is unique)

Mathematical reasoning w.l.o.g.

Revisionism

Few sources treat w.l.o.g. formally, we don't encompass all extant uses

Use our w.l.o.g. principle for **taxonomy**:

- ✓ Dijkstra'95 [EWD1223, not mechanised]
- ✓ HOL WLOG tactics [Harrison'09]
- ✗ Rocq SSReflect
- ✓ Agda Stdlib
- ✗ Lean

Harrison's w.l.o.g. tactics

Harrison's position

Is w.l.o.g. an independent principle or a shorthand?

Implications for **formalisation**:

- ▶ w.l.o.g. as a **shorthand: reconstruct** omission
Harrison posits this view as: ugly/clumsy, inefficient, unfaithful
- ▶ w.l.o.g. **principle: capture** argument structure

Harrison posits a principle is more faithful, and formalises several specialised w.l.o.g. principles, e.g.⁸ by proving all six cases:

$$\frac{\begin{array}{l} \forall x, y, z : (0, \infty). P(x, y, z) \Rightarrow P(y, x, z) \wedge P(x, z, y) \\ \forall x, y, z : (0, \infty). x \leq y \wedge y \leq z \Rightarrow P(x, y, z) \end{array}}{\forall x, y, z : (0, \infty). P(x, y, z)} \quad (\text{wlog}_3)$$

⁸Available here: <https://github.com/jrh13/hol-light/blob/master/int.ml#L612> .

Harrison's w.l.o.g. tactics with symmetric programming

Splitting hairs

Harrison's ($wlog_3$) proof amounts to **shorthand**:

repeats the abstract proof for P six times for each input permutation.

Symmetric programming confines combinatorics to the permutation group S_3

Generating set $S \subseteq \underline{G}$

E.g.: $\{1 \leftrightarrow 2, 2 \leftrightarrow 3\}$ generates S_3

when: $\forall H \subseteq G$ sub-group. $S \subseteq \underline{H} \implies \underline{H} = \underline{G}$

Proposition (S generate G , indexed G -action Γ , proposition $\Gamma \vdash \varphi : \text{Prop}$)

$$\frac{g : S, \gamma : \Gamma \vdash \varphi [\gamma \mapsto g \circ \gamma] \iff \varphi}{\Gamma \vdash \varphi : \text{Prop is equivariant}}$$

Harrison's w.l.o.g. tactics with symmetric programming

$$\frac{\begin{array}{l} (1) \quad \forall x, y, z : (0, \infty). P(x, y, z) \Rightarrow P(y, x, z) \wedge P(x, z, y) \\ (2) \quad \forall x, y, z : (0, \infty). x \leq y \wedge y \leq z \Rightarrow P(x, y, z) \end{array}}{\forall x, y, z : (0, \infty). P(x, y, z)} \quad (\text{wlog}_3)$$

- ▶ $\Gamma := (x, y, z) : (0, \infty)^3$
- ▶ Premise (1) is premise in Proposition $\Rightarrow \gamma : \Gamma \vdash P\gamma : \text{Prop}$ equivariant
 $\Rightarrow$ indexed action of $\mathbf{S}_3$ on $\gamma : \Gamma \vdash P\gamma$
- ▶ Premise (2) is core function $p : (\gamma : \text{Can } c) \rightarrow P\gamma$ for sorting canoniser $c : \underline{\Gamma} \rightarrow \underline{\mathbf{S}}_3$
- ▶ Combinatorics-free proof: $\text{wlog}_3 := \text{wlog}(c, p)(x, y, z)$

Proposition (S generate G , indexed G -action Γ , proposition $\Gamma \vdash \varphi : \text{Prop}$)

$$\frac{g : S, \gamma : \Gamma \vdash \varphi [\gamma \mapsto g \circ \gamma] \Leftarrow \varphi}{\Gamma \vdash \varphi : \text{Prop} \text{ is equivariant}}$$

Harrison's w.l.o.g. tactics with symmetric programming

Harrison's w.l.o.g. tactics are $wlog(c, p)$

by similar analysis, for suitable canoniser c and core function f

One exception⁹

$$\frac{\forall(\mathbf{x}, \mathbf{y}) : \mathbb{R}^2. P(\mathbf{x}, \mathbf{y}, 0)}{\forall(\mathbf{x}, \mathbf{y}, \mathbf{z}) : \mathbb{R}^3. \mathbf{z} = 0 \implies P(\mathbf{x}, \mathbf{y}, \mathbf{z})} \quad (\text{pad2d3d_tac})$$

(This presentation de-emphasises importance for automated goal-simplification.)

Takeaway

Revisionist wlog as crisp taxonomy for extant w.l.o.g. arguments

⁹Available here: <https://github.com/jrh13/hol-light/blob/master/int.ml#L612> .

Harrison's concluding remarks

Harrison connects his w.l.o.g. principles to symmetry, but not to group actions.

Harrison considers **domain-specific reflective libraries** for w.l.o.g. reasoning, such as ones for geometric reasoning. He argues they are **too specific**.

Groups actions maintain a flexible level of generality, and high expressivity.

Dijkstra's w.l.o.g. misery

Note EWD1223

presents a general w.l.o.g. principle for binary relations, with two goals:

- ▶ explicate it formally
- ▶ kvetch—using unordered pairs—about how w.l.o.g. was explained to him

Dijkstra's w.l.o.g. principle

$\varphi, \psi : A^2 \rightarrow \mathbf{Prop}$ predicates on ordered pairs satisfying:

1. $(x, y) : A^2 \vdash \varphi(x, y) = \varphi(y, x)$ and similarly for ψ .
2. finding a third predicate $\xi : A \rightarrow \mathbf{Prop}$ such that $(x, y) : A^2 \vdash \varphi(x, y) \wedge \xi x \implies \psi(x, y)$.
3. showing that $\varphi(x, y) \implies (\xi x \vee \xi y)$;

concluding that, w.l.o.g., $\varphi(x, y) \implies \psi(x, y)$.

Dijkstra's w.l.o.g. principle as a $\text{wlog}(c, p)$

$(x, y) : A^2 \vdash \varphi, \psi : \text{Prop}$ predicates on ordered pairs satisfying:

1. φ, ψ invariant $\implies \mathbf{S}_2$ action on them
3. predicate $\xi : A \rightarrow \text{Prop}$ and proof $q : \varphi(x, y) \implies (\xi x \vee \xi y)$ provide a canoniser:

$$(x, y) : A^2 \vdash c : \varphi(x, y) \rightarrow \underline{\mathbf{S}}_2 \quad cp := \begin{cases} \xi x : & \mathbf{I} \\ \xi y : & \dashv \end{cases} \quad p \in \text{Can } c \iff \varphi(x, y) \wedge \xi x$$

2. finding a core function $(x, y) : A^2 \vdash f : \text{Can } c \rightarrow \psi(x, y)$.

We can now prove: $(x, y) : A^2 \vdash \text{wlog}(c, f) : \varphi(x, y) \implies \psi(x, y)$.

Misery of the unordered pair

Dijkstra recounts: when the w.l.o.g. argument was presented/justified to him informally, the presenter included this step:

4. 'by similar reasoning, $(x, y) : A^2 \vdash \varphi(x, y) \wedge \xi y \implies \psi(x, y)$ '

Dijkstra argues: replacing φ, ψ with **unordered pairs** predicates:

$$\Phi : (A^2 / (x, y) \sim (y, x)) \rightarrow \text{Prop}$$

step (4) coincides with step (2), hence unnecessary.

Arguing what Dijkstra must or may not be forced to concede is in the realm of **Philosophy of Mathematics**—out of scope.

Some observations

- **quotients are subtle**, esp. in the presence of multiple symmetries. Compare:

let S, T be two equilateral triangles in the plane, then S and T may be different

vs

let S, T be two equilateral triangles in the plane, each quotiented by affine symmetries. Then $S = T$.

NB: a standard subtlety when working with quotients.

- '(2) and (4) **coincide**' means '**judgemental** equality of symmetric unordered pairs' McBride's **Ripley phenomena**¹⁰ kicks-in causing **undedicable type-checking**.

NB: may be only a complication due to the dependently-typed metalanguage

¹⁰See other Ripley phenomena in McBride's note: <http://strictlypositive.org/Ripley.pdf>.

Dijkstra's w.l.o.g. misery

Arguing what Dijkstra must or may not be forced to concede is in the realm of **Philosophy of Mathematics**—out of scope.

Some observations

- Dijkstra's misery may be a **strawman**
Whoever presented w.l.o.g. to Dijkstra did **not explicate** the formal argument.

Relevant **symmetry**, predicate **equivariance** and **canoniser** left implicit.

Presenter may have **demonstrated** how the symmetric argument looks like.
Necessary repetition to re-construct implicit data.

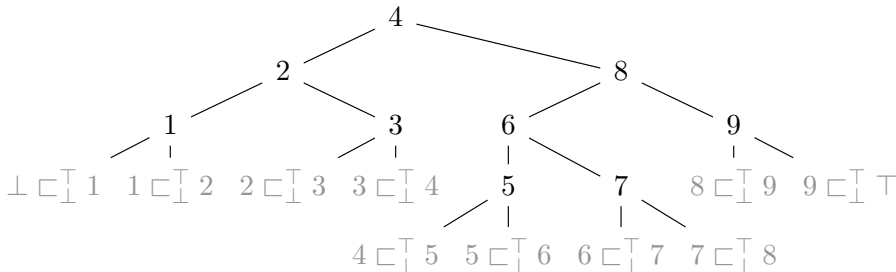
Once symmetry, equivariance, and canoniser are explicit, proving the symmetric case indeed becomes unnecessary, but not before!

Talk structure

- ▶ Introduction
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 - ▶ Indexed data-types
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McBride's binary search trees [McBride'14]

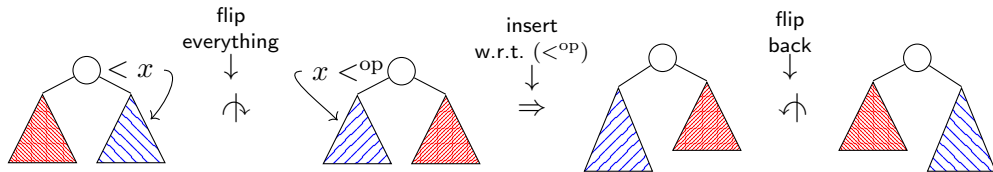
$$\frac{p : a \sqsubset_{\perp}^{\top} b}{\text{Leaf}_p : \mathbb{T}_{(\sqsubset)}(a, b)} \quad \frac{\ell : \mathbb{T}_{(\sqsubset)}(a, x) \quad r : \mathbb{T}_{(\sqsubset)}(x, b)}{\text{Node} \langle x, \ell, r \rangle : \mathbb{T}_{(\sqsubset)}(a, b)}$$



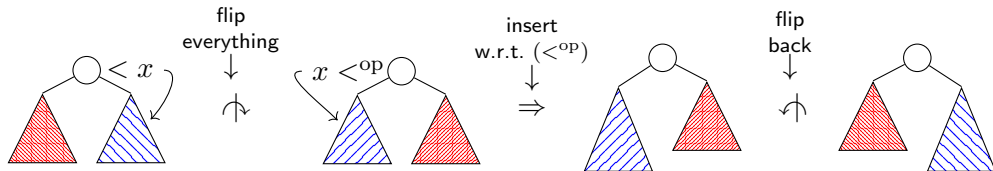
proofs stored in **leafs** keep the **nodes in order**.

our goal: define insertion without the **symmetric case**

$$(\sqsubset) : \mathbf{Total}_{\mathbb{K}}, k : \mathbb{K} \vdash \text{insert} : ((x, y) : \text{interval}_{\mathbb{K}}) \rightarrow x \sqsubset_{\perp}^{\top} k \sqsubset_{\perp}^{\top} y \rightarrow \mathbb{T}_{(\sqsubset)}(x, y) \rightarrow \mathbb{T}_{(\sqsubset)}(x, y)$$



Tree.lean



Tree.lean

Challenges:

- ▶ Symmetry compatible induction principles
- ▶ We'll develop a theory of data-types with symmetry
augments usual theory of data-types with group actions.

Data-types with symmetry

Plan

- ▶ recap: semantic structure for **simply-typed** data-types
- ▶ **simply-typed** data-types with symmetry
- ▶ recap: **indexed** data-types
- ▶ **indexed** data-types with symmetry

Recap: semantic structure for simply-typed data-types

data types	top-level constructor + sub-structure	example structural induction
$\text{List } A = \text{Nil}$ $\quad \{ (::) \} \times A \times \text{List } A$	$X \mapsto \text{Nil}$ $\quad \{ (::) \} \times A \times X$	length: $\begin{cases} \text{Nil} & \mapsto 0 \\ ((::), a, n) & \mapsto 1 + n \end{cases}$
$\text{Tree } A = \text{Leaf}$ $\quad \{ \text{Node} \} \times \text{Tree } A$ $\quad \quad \times A \times \text{Tree } A$	$X \mapsto \text{Leaf}$ $\quad \{ \text{Node} \} \times X$ $\quad \quad \times A \quad \times X$	depth: $\begin{cases} \text{Leaf} \mapsto 0 \\ \langle \text{Node}, m, x, k \rangle \\ \quad \mapsto 1 + \max \{ m, k \} \end{cases}$

etc. for strictly positive data-types

Recap: semantic structure for simply-typed data-types

General picture:

data type + constructors	top-level constructor + sub-structure	structural induction
	functor $F : \mathbf{Set} \rightarrow \mathbf{Set}$ FX : constructors available using substructures $x \in X$	
initial algebra $\mathbf{roll} : F(\mu F) \xrightarrow{\cong} \mu F$ constructs data from constructor + data for substructures evaluates each constructor homomorphically using algebra structure:		algebra $f : FA \rightarrow A$ evaluate constructor given values for sub-structures
	$\frac{f : FA \rightarrow A}{\mathbf{fold} \ f : (\mu F, \mathbf{roll}) \rightarrow (A, f)}$	F -algebra F -algebra homomorphism

Simply-typed data-types with **symmetry**

Examples

- ▶ Pointwise symmetry on lists (G -action A):

$$\boxed{\text{Nil}} \xrightarrow{g \circledast} \boxed{\text{Nil}} \qquad \boxed{a :: x} \xrightarrow{g \circledast} \boxed{(g \circledast a) :: (g \circledast x)}$$

- ▶ Tree flipping symmetry (S_2 -action A):

$$\boxed{\text{Leaf}} \xrightarrow{g \circledast} \boxed{\text{Leaf}} \qquad \begin{array}{ccc} & \boxed{n} & \\ x & / \quad \backslash & y \end{array} \xrightarrow{\text{flip} \circledast} \begin{array}{ccc} & \boxed{n} & \\ \text{flip} \circledast y & / \quad \backslash & \text{flip} \circledast x \end{array}$$

Simply-typed data-types with **symmetry**

Symmetry strength (G group, $F : \mathbf{Set} \rightarrow \mathbf{Set}$ functor)

natural transformation

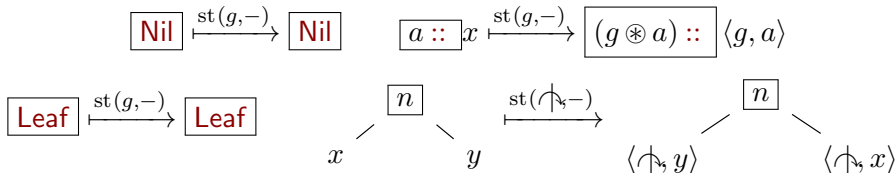
$$\text{st}_X : \underline{G} \times FX \rightarrow F(\underline{G} \times X)$$

satisfying:

$$t : FX \vdash \text{st}_X(e, t) = \text{map}(x \mapsto \langle e, x \rangle) t$$

$$g, h : \underline{G}, t : FX \vdash \text{st}_X(g * h, t) = \text{map}(\langle g', h', x \rangle \mapsto \langle g' * h', x \rangle) (\text{st}_{\underline{G} \times X}(g, \text{st}_X(h, t)))$$

Examples



Simply-typed data-types with **symmetry**

Symmetry strength (G group, $F : \mathbf{Set} \rightarrow \mathbf{Set}$ functor)

natural transformation

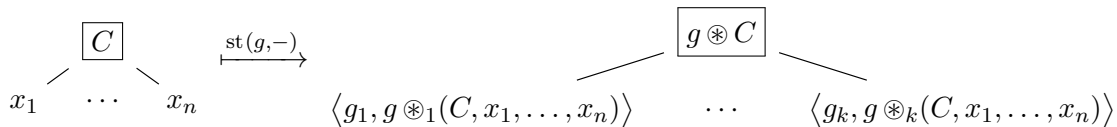
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Generally



Simply-typed data-types with **symmetry**

Symmetry strength (G group, $F : \mathbf{Set} \rightarrow \mathbf{Set}$ functor)

natural transformation

$$\mathrm{st}_X : \underline{G} \times FX \rightarrow F(\underline{G} \times X)$$

satisfying:

$$t : FX \vdash \mathrm{st}_X(e, t) = \mathbf{map}(x \mapsto \langle e, x \rangle) t$$

$$g, h : \underline{G}, t : FX \vdash \mathrm{st}_X(g * h, t) = \mathbf{map}(\langle g', h', x \rangle \mapsto \langle g' * h', x \rangle) (\mathrm{st}_{\underline{G} \times X}(g, \mathrm{st}_X(h, t)))$$

Unit law

$$\begin{array}{ccc} \begin{array}{c} \boxed{C} \\ \swarrow \quad \searrow \\ x_1 \quad \cdots \quad x_n \end{array} & \xrightarrow{\mathrm{st}(e, -)} & \begin{array}{c} \boxed{C} \\ \swarrow \quad \searrow \\ \langle e, x_1 \rangle \quad \cdots \quad \langle e, x_n \rangle \end{array} \end{array}$$

Simply-typed data-types with **symmetry**

Symmetry strength (G group, $F : \mathbf{Set} \rightarrow \mathbf{Set}$ functor)

natural transformation

$$\text{st}_X : \underline{G} \times FX \rightarrow F(\underline{G} \times X)$$

satisfying:

$$t : FX \vdash \text{st}_X(e, t) = \text{map}(x \mapsto \langle e, x \rangle) t$$

$$g, h : \underline{G}, t : FX \vdash \text{st}_X(g * h, t) = \text{map}(\langle g', h', x \rangle \mapsto \langle g' * h', x \rangle) (\text{st}_{\underline{G} \times X}(g, \text{st}_X(h, t)))$$

Multiplication law

$$\begin{array}{c} \boxed{C} \xrightarrow{\text{st}(h, -)} \boxed{h \otimes C} \xrightarrow{\text{map}(\langle (*), - \rangle \circ \text{st}(g, -))} \boxed{(g * h) \otimes C} \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \quad \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ x_1 \quad \cdots \quad x_n \langle h_1, y_1 \rangle \quad \cdots \quad \langle h_k, y_k \rangle \langle (g * h)_1, z_1 \rangle \quad \cdots \quad \langle (g * h)_k, z_k \rangle \end{array}$$

A G -strength **lifts** G -actions from A to $F\underline{A}$

Proposition

For every symmetry strength st and every G -action A , there is a G -action $F_{\text{st}}A$ with $\underline{F_{\text{st}}A} := F\underline{A}$ and

$$\left(\overset{F_{\text{st}}A}{*} \right) : \underline{G} \times F\underline{A} \xrightarrow{\text{st}} F(\underline{G} \times \underline{A}) \xrightarrow{\text{map}(\overset{A}{*})} F\underline{A}$$

Moreover, if $h : A \circledast \rightarrow B$ is equivariant then so is $\text{map}h : F_{\text{st}}A \circledast \rightarrow F_{\text{st}}B$.

Two S_2 -strengths on $\text{Tree } \mathbb{N}$ induce two different actions

Flipping the tree

$$\text{st}^{\text{Tree } \mathbb{N}; S_2} \langle F, \text{Node } \ell n r \rangle := \text{Node } \langle F, r \rangle n \langle F, \ell \rangle$$

$$F \circledast \left(\begin{array}{c} & & 3 \\ & \swarrow & \searrow \\ 1 & 2 & 4 \end{array} \right) = \left(\begin{array}{c} 3 & & \\ \swarrow & & \searrow \\ 4 & 2 & 1 \end{array} \right)$$

Flipping the top node

$$\text{st}'^{\text{Tree } \mathbb{N}; S_2} \langle F, \text{Node } \ell n r \rangle := \text{Node } \langle I, r \rangle n \langle I, \ell \rangle$$

$$F \circledast' \left(\begin{array}{c} & & 3 \\ & \swarrow & \searrow \\ 1 & 2 & 4 \end{array} \right) = \left(\begin{array}{c} 3 & & \\ \swarrow & & \searrow \\ 4 & 2 & 1 \end{array} \right)$$

An equivariant algebra is **both an F -algebra and equivariant**

Definition

An **equivariant $\langle F, \text{st} \rangle$ -algebra** $\langle A, f \rangle$ consists of a G -action A and an equivariant F -algebra $f : F_{\text{st}} A \circlearrowright A$, i.e., an F -algebra $f : F\underline{A} \rightarrow \underline{A}$ such that

$$f(g \overset{F_{\text{st}} A}{\circlearrowright} t) = g \overset{A}{\circlearrowright} f(t)$$

$$\begin{array}{ccccc} \underline{G} \times F\underline{A} & \xrightarrow{\text{st}} & F(\underline{G} \times \underline{A}) & \xrightarrow{F(\circlearrowright)} & F\underline{A} \\ \text{id} \times f \downarrow & & = & & \downarrow f \\ \underline{G} \times \underline{A} & \xrightarrow{(\circlearrowright)} & & & \underline{A} \end{array}$$

An **equivariant algebra homomorphism** $h : \langle A, f \rangle \circlearrowright \langle B, g \rangle$ is simultaneously an F -homomorphism and equivariant $h : A \circlearrowright B$.

The initial F -algebra is the **initial equivariant algebra**

For every F -algebra $f : FX \rightarrow X$ there is a unique **parameterised homomorphism** $\mathbf{pfold} f : \underline{G} \times \mu F \rightarrow X$ satisfying

$$\mathbf{pfold} f \langle g, \mathbf{roll} t \rangle = f(\mathbf{map}(\mathbf{st}(g, t)))$$

Theorem

Let $\mathbf{roll} : F\mu F \rightarrow \mu F$ be an initial algebra for F with G -strength $\mathbf{st}$. Defining

$$(\circledast) := \mathbf{pfold} \mathbf{roll} : \underline{G} \times \mu F \rightarrow \mu F$$

makes $\langle \mu F, \mathbf{roll} \rangle$ into the **initial** equivariant $\langle F, \mathbf{st} \rangle$ -algebra: for every equivariant algebra $f : F_{\mathbf{st}} A \circledast \rightarrow A$, the unique F -homomorphism $\mathbf{fold} f : \mu F \circledast \rightarrow A$ is equivariant.

Talk structure

- ▶ Introduction
- ▶ Remedial group theory basics
- ▶ Simply-typed symmetric programming
- ▶ Dependently-typed symmetric programming
 - ▶ Indexed actions and w.l.o.g.
 - ▶ Mathematical reasoning: Dijkstra'95 & Harrison'09
- ▶ Symmetric user-defined data-types
 - ▶ Simply-typed data-types
 - ▶ Indexed data-types
- ▶ Equivariant reasoning

We want core functions to be **equivariant where defined**

Assume an equivariant $\varphi : A \circledast \rightarrow B$ extends a core function $f : \text{Can } c \rightarrow \underline{B}$. Then:

$$\begin{array}{ccccc} \varphi \text{ extends } f & \varphi \text{ is equivariant} & \varphi \text{ extends } f & & \\ \downarrow & \downarrow & \downarrow & & \\ f(g \circledast x) & = \varphi(g \circledast x) & = g \circledast \varphi(x) & = g \circledast fx \end{array}$$

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Let A, B be G -actions and $X \subseteq \underline{A}$. Then, $f : X \rightarrow \underline{B}$ is **equivariant** when

$$\forall g \in \underline{G}. \forall x \in X. (g \overset{A}{\circledast} x \in X \implies f(g \overset{A}{\circledast} x) = g \overset{B}{\circledast} fx)$$

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Theorem

Let A, B be G -actions, $c : \underline{A} \rightarrow \underline{G}$ a canoniser, $f : \text{Can } c \rightarrow \underline{B}$ an equivariant core function. Then $\text{wlog}(c, f) : A \circlearrowright B$ is the unique equivariant function extending f .

Are canonisers **equivariant**?

$$(ca * g^{-1}) \circledast (g \circledast a) = ca \circledast a \in \text{Can } c$$

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A canoniser $c : \underline{A} \rightarrow \underline{G}$ is **strictly equivariant** when

$$c(g \circledast a) = ca * g^{-1} = g \overset{G^{-1}}{\circledast} (ca)$$

i.e. when $c : A \circledast \rightarrow G^{-1}$.

Example

Strict equivariance is a **strong assumption** and a **strong restriction**

Theorem

A canoniser $c : \underline{A} \rightarrow \underline{G}$ is strictly equivariant if and only if, for every G -action B , every core function $f : \text{Can } c \rightarrow \underline{B}$ is equivariant.

Strict equivariance is a **strong assumption** and a **strong restriction**

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Proposition

If A has a strictly equivariant canoniser then A is **semi-regular**:

$$g \circledast a = a \implies g = e$$

Example

C_4 -action on Board_n is not semi-regular, but it is on $(\text{Dir} \times \text{Board}_n)$. □

The issue is in **indistinguishability** of symmetries

Definition

For a G -action A and $a \in \underline{A}$, the **indistinguishability** relation is

$$g \stackrel{a}{\sim} h \iff g \circledast a = h \circledast a$$

The quotient $(G/A)_a := G/\stackrel{a}{\sim}$ carries an A -indexed G -action:

$$g \circledast_a [h]_{\stackrel{a}{\sim}} := [h * g^{-1}]_{g \circledast_a}$$

Side note: **orbit-stabiliser theorem** is an **equivariant isomorphism**

$$\begin{aligned} G \circledast [a] &:= \{h \circledast a \mid h \in \underline{G}\} \\ g \circledast_a (h \circledast a) &:= h \circledast a = (h * g^{-1}) \circledast (g \circledast a) \in G \circledast [g \circledast a] \end{aligned}$$

Theorem (Orbit-Stabiliser)

Let A be a G -action. There is an equivariant isomorphism:

$$\varphi_a : (G/A)_a \xrightarrow[\circledast]{\cong} G \circledast [a] \quad \varphi_a[h]_{\sim} := h \circledast a$$

Collapsing indistinguishable symmetries **weakens** strict equivariance

A canoniser $c : \underline{A} \rightarrow \underline{G}$ is **equivariant** when $(a \mapsto [c a]_a) : A \twoheadrightarrow G/A$.

Example

$F \circledast \langle a, a \rangle = \langle a, a \rangle$, so $I \stackrel{\langle a, a \rangle}{\sim} F$, thus

$$c(F \circledast \langle a, a \rangle) = I \stackrel{\langle a, a \rangle}{\sim} F = F \circledast_{\langle a, a \rangle} c \langle a, a \rangle$$



Theorem

Using the axiom of choice, every G -action A has an equivariant canoniser $c : \underline{A} \rightarrow \underline{G}$.

Equivariant canonisers make every **stable** core function equivariant

Every equivariant f satisfies $g \in G_a \implies g \circ f\ a = f(g \circ a) = f\ a$, so

$$G_a \subseteq G_{f\ a}$$

A

core function $f : \text{Can } c \rightarrow \underline{B}$ is **stable** when $\forall a \in \text{Can } c. G_a \subseteq G_{f\ a}$.

Theorem

c is equivariant iff every stable core function $f : \text{Can } c \rightarrow \underline{B}$ is equivariant.

Symmetric programming

Exploit symmetries for computation and reasoning—**avoid symmetric cases**.

Mathematical foundations for symmetric programming semantics:

- ▶ **Abstractions** for symmetric programming
- ▶ **Simply-typed** reasoning about equivariance
- ▶ **Dependently-typed** and mathematical reasoning
- ▶ User-defined **data-types** through initial algebra semantics and induction principles
 - ▶ Simply-typed data-types
 - ▶ Dependently-typed/indexed data-types

Next time, on **WG 2.1**...

...for you in the paper

- ▶ more examples (DAGs, BSTs)
- ▶ indexed datatypes
- ▶ Harrison's tactic (non)-examples
- ▶ EWD1223

...for us still in the works

- ▶ more examples (???)
- ▶ mechanisation
- ▶ programming abstractions
- ▶ efficient implementation